

Determining critical phase transition temperature in an Ising model simulation

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Room for abstract.

I. INTRODUCTION

A. The Ising model

The Ising model named after the German physicist Erns Ising is a model in statistical mechanics. The model consists of discrete magnetic spin dipole moments that take values ± 1 ("up" and "down" respectively) arranged on a lattice. For the purposes of our simulation we used a square lattice. The spins interact with neighbouring spins. In our implementation we consider the von Neumann neighbourhood of only 4 neighbouring spins in the $\pm x$ and $\pm y$ directions. The model exhibits phase transitions that can be associated with the gas-liquid transition or spontaneous magnetisation.

B. Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition. It is know from theory that under a certain critical temperature the model stabilises into completely up or down state. Conversely, above this critical, on average, the model remains a random mixture of up and down spins.

II. THEORY

Theory.

III. METHODOLOGY

A. Metropolis algorithm

TODO: <Explain metropolis algorithm>

B. Determining parameters for final simulation

TODO: <Explain how we arrived at simulation parameters>

IV. RESULTS

TODO: <Give results>

TODO: <Give figures of results>

V. COMPARISON TO THEORY

A. Theoretical solution for the critical phase transition temperature

TODO: <Give derivation of critical temperature in the Ising model>

B. Comparison of theory and our results

TODO: <Compare our results with theoreticla values>

VI. CONCLUSIONS

Conclusion.

[1] F. Chen, M.C. Cheung, P.M. Sweeney, W.D. Kirkey, M. Furis, and A.N. Cartwright. J. Appl. Phys. **93**, 4933 (2003).